

School of Computer Science and Engineering

B.Tech (CSE) 7 Semester

Internship Title: Sign Language Recognition using Sign-IDD Model

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Organization: CSIR – 4PI (CSIR–Fourth Paradigm Institute)

Internship Duration: April 2026 – July 2026

Internship Overview

- **Program:** CSIR – 4PI (CSIR–Fourth Paradigm Institute)
- **Domain:** Computer Science and Engineering
- **Duration:** April 2026 – July 2026
- **Project:** Language Recognition using Sign-IDD Model
- **Programming Language:** Python
- **Institution:** IILM University, Greater Noida
- **Academic Session:** 2026–27

Introduction

- Sign language is an important mode of communication for hearing- and speech-impaired individuals.
- This project focuses on recognizing **Indian Sign Language (ISL)** gestures from video sequences.
- The **Sign-IDD dataset** was used as the source of labeled ISL gesture videos.
- **OpenPose** was used to extract 2D body and hand keypoints from video frames.
- An **LSTM-based deep learning model** was developed to learn temporal gesture patterns and predict gesture labels.
- The system aims to reduce communication barriers and support hearing-impaired individuals through AI-based gesture recognition.

Objectives

- To develop a **machine-learning-based system** for recognizing Indian Sign Language gestures.
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- To study and organize the **Sign-IDD dataset** containing labeled ISL gesture videos.
- To preprocess videos using **FFmpeg** for consistent format and frame rate.
- To extract **2D body and hand keypoints** using OpenPose.
- To generate structured `.skel files` containing skeletal data.
- To normalize skeletal data and prepare **temporal sequences**.
- To develop an **LSTM-based deep learning model** for gesture classification.
- To test the system and evaluate its **gesture recognition performance**.

Methodology

- **Dataset Preparation** – Organize the Sign-IDD dataset according to ISL gesture classes.
- **Video Preprocessing** – Convert videos into a consistent format using **FFmpeg**.
- **Skeleton Extraction** – Extract 2D body and hand keypoints using **OpenPose** and store them in .skel files.
- **Feature Processing** – Normalize skeleton coordinates and extract temporal motion features.
- **Model Development** – Develop an **LSTM-based deep learning model** using PyTorch.
- **Training & Evaluation** – Divide the dataset into training, validation and testing sets and evaluate the model.
- **Testing** – Process new videos and generate the predicted gesture label.

Technologies Used in Internship

- **Python** – Data processing and model development
- **OpenPose** – 2D body and hand keypoint extraction
- **Sign-IDD Dataset** – Labeled ISL gesture videos
- **FFmpeg** – Video preprocessing and conversion
- **PyTorch** – LSTM model development and training
- **NumPy** – Numerical and skeletal data processing
- **OpenCV** – Video and image processing
- **LSTM** – Learning temporal patterns in gestures
- **.skel Files** – Storing frame-wise 2D skeletal keypoints

Results & Future Scope

- LSTM model gave **stable predictions** for most ISL gestures.
- Average processing time was **1.5–2 minutes per video** on CPU.
- Use **GPU acceleration and 3D pose estimation** for faster and better recognition.
- Add **real-time webcam, larger datasets, and sentence-level recognition**.

Conclusion

- Successfully developed an **Indian Sign Language (ISL) recognition system** using the Sign-IDD dataset.
- **OpenPose** was used to extract 2D skeletal keypoints and **LSTM** was used for gesture classification.
- The system produced **stable and consistent predictions** during testing.
- The project demonstrates that combining **skeletal keypoint extraction with temporal deep learning** is effective for ISL gesture recognition

Thank You